NGUYEN TAN THAI HUNG

ntthung@gmail.com
people.sutd.edu.sg/~ntthung
github.com/ntthung
+65 9159 7257

Paleoclimate
Water resources management
Statistics & Machine learning
R • C++ • MATLAB • Python

Research statement

I study variability and changes in the water cycle across spatiotemporal scales, and apply this knowledge to water resources management. I advocate open science and science communication.

Education

2016 – 2021	**Singapore University of Technology and Design (SUTD)** Doctor of Philosophy	Singapore
2009 – 2012	**Delft University of Technology (TUDelft)** **National University of Singapore (NUS)** Double Master of Science degrees	The Netherlands Singapore
2005 – 2009	**National University of Singapore** **(jointly conferred by the University of Melbourne)** Bachelor of Engineering (Civil)	Singapore Australia

Awards and scholarships

2019	Lamont Postdoctoral Fellowship, Columbia University Teaching Excellence Award–Most Outstanding Teaching Assistant, SUTD Best Research Project, SUTD Research Fest
2018	Winner, SUTD 3-Minute Thesis Competition
2017	Outstanding Teaching Assistant Award, SUTD
2016	President's Graduate Fellowship, SUTD
2010	Keppel Corporation Study Grant, SGD 30,000
2009	Singapore-Delft Water Alliance Study Grant, SGD 50,000
2009	Engineering Innovation & Research Award, NUS
2007	NUS Award for Studying Abroad, SGD 11,000
2005	Civil Engineering Scholarship, SGD 24,000

Teaching

2020	**Teaching Associate, Columbia University** ***(invited, remotely)*** Course: Predicting the Effects of Climate Change on Global Forests
2016 – 2018	**Graduate Teaching Assistant, SUTD** Courses: Engineering Systems Architecture, Quality and Reliability

Talks

Academic		
2019	*Streamflow variability in Monsoon Asia over the past eight centuries*	
	Woods Hole Oceanographic Institution	USA
	Tree Ring Lab, Lamont Doherty Earth Observatory	USA
	Earth Observatory of Singapore & Asian School of Environment, Nanyang Technological University	Singapore
	Centre for Climate Research Singapore	Singapore
2018	*A linear dynamical systems approach to streamflow reconstruction*	
	Centre for Climate Research Singapore	Singapore
Industrial		
2018	*Building trust: An engineer's non-engineering challenge*	
	Singapore University of Technology and Design	

Conferences and workshops

AGU 2020 (American Geophysical Union)	Virtual
MODSIM 2019 (Modelling & Simulation Society of Australia and New Zealand)	Australia
AOGS 2019 (Asia Oceania Geosciences Society)	Singapore
EGU 2019 (European Geosciences Union)	Austria
SAPEC-2019 Isotopes in the Environment	Vietnam
AGU 2018	USA
29th European Dendroecological Field Week, 2018	Croatia
EGU 2018	Austria
AOGS 2017	Singapore

Service

Conferences and field school organized		
2020	**SAPEC-2020 Data and Methods in Paleo Proxy Sciences** *Organizing team member* Field school with 40 scientists and students from 12 countries.	Vietnam
2018 – 2019	**ESD Summer Conference Series** *Founder and lead organizer* Student-led annual conferences showcasing research in Engineering Systems & Design	Singapore
Peer review		
Since 2017	Reviewer for *Quaternary Science Review, Water Resources Research, Journal of Water Resources Planning and Management, Environmental Modelling & Software*, and *Water Management*.	

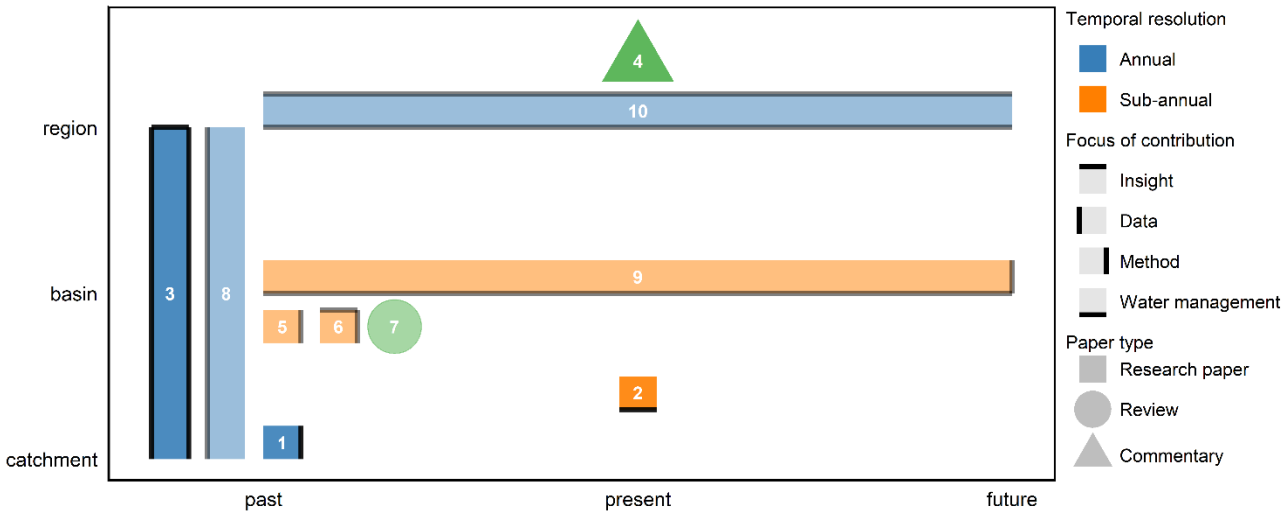
Community

Since 2019 Regional Representative, Past Global Changes (PAGES) Early Career Network.

Publications (published and planned)

Overview

Below is a layout of my publications and projects in pipeline. The numbers follow the list in the next section.



Journal papers

1. **Nguyen, H. T. T.**, & Galelli, S. (2018). A Linear Dynamical Systems Approach to Streamflow Reconstruction Reveals History of Regime Shifts in Northern Thailand. *Water Resources Research*, 54(3), 2057–2077. DOI: 10.1002/2017WR022114
2. Libisch-Lehner, C. P., **Nguyen, H. T. T**., Taormina, R., Nachtnebel, H. P., & Galelli, S. (2019). On the Value of ENSO State for Urban Water Supply System Operators: Opportunities, Trade-Offs, and Challenges. Water Resources Research, 55(4), 2856–2875. DOI: 10.1029/2018WR023622
3. **Nguyen, H. T. T.**, Turner, S. W. D., Buckley, B. M., & Galelli, S. (2020). Coherent Streamflow Variability in Monsoon Asia Over the Past Eight Centuries—Links to Oceanic Drivers. Water Resources Research, 56(12). DOI: 10.1029/2020WR027883
4. Galelli, S., **Nguyen, H. T. T.**, Turner, S. W. D., and Buckley, B. M. (2021). Time to use dendrohydrological data in water resources management? Commentary, to appear in *Journal of Water Resources Planning and Management*.
5. **Nguyen, H. T. T.**, Galelli, S., Xu, C., & Buckley, B. (2020). Multi-Proxy, Multi-Season Streamflow Reconstruction with Mass Balance Adjustment. Earth and Space Science Open Archive. DOI: 10.1002/essoar.10504791.1

6. Chowdhury, A. F. M. K., Dang, T. D., **Nguyen, H. T. T.**, Koh, R., & Galelli, S. (2021). The Greater Mekong's climate-water-energy nexus: how ENSO-triggered regional droughts affect power supply and CO2 emissions. Accepted in Earth's Future. DOI: 10.1002/essoar.10504393.1

In pipeline

7. Buckley, B.M., Nguyen, H., Galelli, S., Ummenhofer, C.C., D'Arrigo, R.D., Hansen, K.G., and Stahle, D.K. The Southeast Asian Dendrochronology Project: Hydroclimate variability over the past millennium. *Quaternary Science Reviews* (invited review, manuscript in preparation).
8. **Nguyen**, **H.,** Barbarossa, V., Turner, S.W.D., Buckley, B.M., and Galelli, S. A high-resolution paleo streamflow record for Monsoon Asia with surrogate streamflow data. Presented at MODSIM 2019; work in progress.
9. **Nguyen**, **H. T. T.,** Vu, D. T., and Galelli, S. On the value of streamflow reconstruction to flood management in the Chao Phraya River Basin. *Work in progress.*
10. Paleoclimate model simulation and paleostreamflow reconstruction. This work will be carried out in my postdoc.

Media

PhD and Parenthood: An Insider's Story.
Blog post on doing a PhD as a parent, first appeared on *The Early Pages,* blog site of the PAGES Early Career Network, and reproduced by TODAYonline (a Singapore newspaper). For this article I was also interviewed live on air by Channel News Asia's FM 93.8.
https://theearlypages.blogspot.com/2019/08/phd-and-parenthood-insiders-story.html

Industrial experience

2011 – 2015 **Keppel Group of Companies**

I led teams to commission, operate, and maintain water treatment plants.

- *Senior Operations & Maintenance Engineer (2015)* — Singapore
  Ulu Pandan NEWater Plant
- *Process Engineer (2013 – 2014)* — Qatar
  Doha North Sewage Treatment Works
- *Trainee (2011 – 2012)* — Singapore
  Keppel Seghers

Awards:

- Safety Supervisor of the year (2014)
- Safety Innovation Project (2014, 2015)